

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df=pd.read_csv("social_media_usage_1000.csv")
print("---Preview of 1st Five Records:---\n",df.head())
print("\n---Diplaying info about Columns and Data Types:---\n")
print(df.info())
```

```
---Preview of 1st Five Records:---
   User_ID  Age  Gender  Instagram_hrs  Facebook_hrs  YouTube_hrs  \
0         1  51.0   Male             0.8             2.1           1.2
1         2  41.0  Female             2.6             2.2           3.5
2         3   NaN   Male             3.5             1.3           3.6
3         4  55.0  Female             3.2             1.4           0.6
4         5  20.0  Female             NaN             0.3           4.4
```

```
   Purpose  Location  Device_Type  Daily_Sessions
0  Education    Urban    Desktop             5
1         NaN  Suburban    Desktop            11
2  Socializing   Rural    Mobile             2
3         Work   Rural    Mobile             1
4  Socializing   Urban    Mobile             2
```

```
---Diplaying info about Columns and Data Types:---
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 10 columns):
#   Column                Non-Null Count  Dtype
---  -
0   User_ID               1000 non-null  int64
1   Age                  950 non-null   float64
2   Gender               950 non-null   object
3   Instagram_hrs        950 non-null   float64
4   Facebook_hrs         950 non-null   float64
5   YouTube_hrs          950 non-null   float64
6   Purpose              950 non-null   object
7   Location             1000 non-null  object
8   Device_Type          1000 non-null  object
9   Daily_Sessions       1000 non-null  int64
dtypes: float64(4), int64(2), object(4)
memory usage: 78.3+ KB
None
```

```
df['Gender'].fillna(df['Gender'].mode()[0],inplace=True)
df['Age'].fillna(df['Age'].median(),inplace=True)
for col in ['Instagram_hrs', 'Facebook_hrs', 'YouTube_hrs']:
    df[col].fillna(df[col].mean(), inplace=True)
df['Purpose'].fillna('Unknown', inplace=True)
```

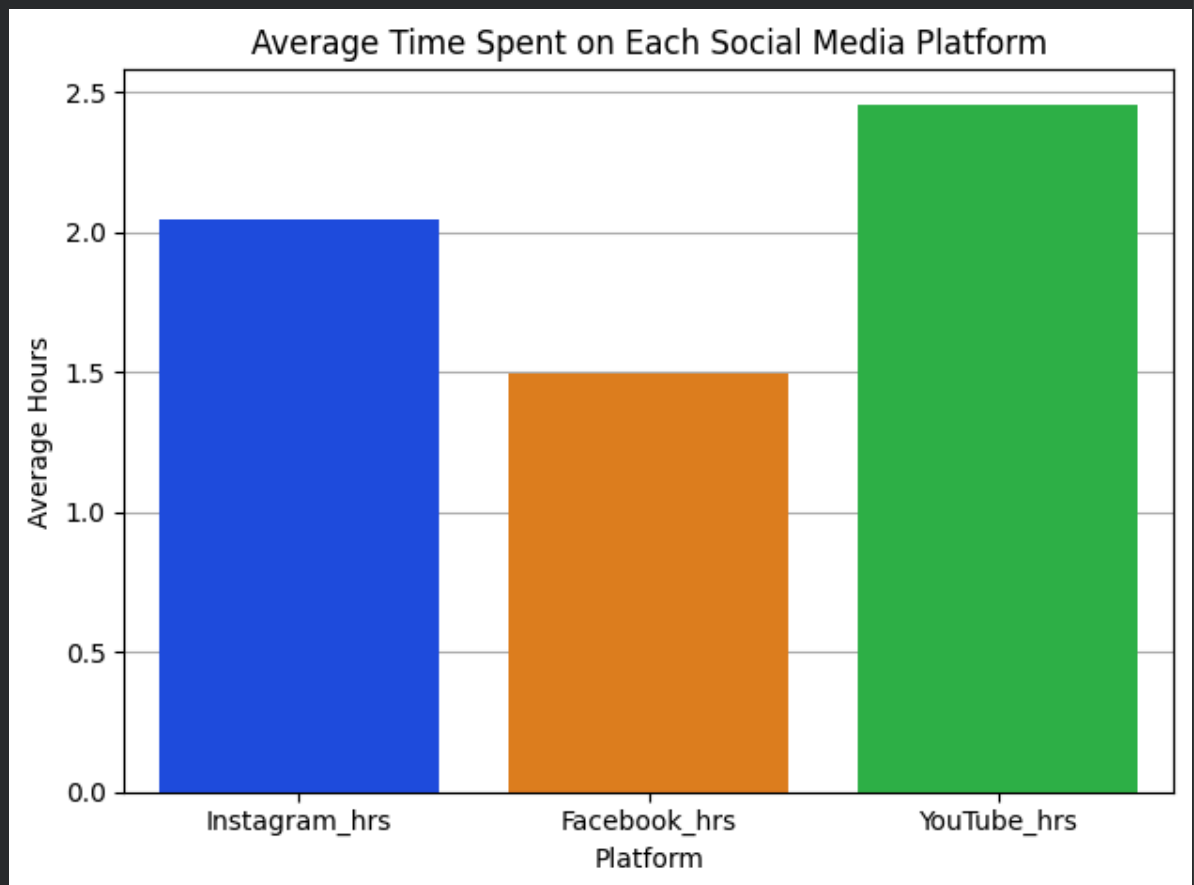
```
/tmp/ipython-input-2766800091.py:4: FutureWarning: A value is trying to be se
The behavior will change in pandas 3.0. This inplace method will never work b
```

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.

```
df[col].fillna(df[col].mean(), inplace=True)
```

```
df['Total_Hours'] = df[['Instagram_hrs', 'Facebook_hrs', 'YouTube_hrs']].sum
```

```
from re import X
mean_platform = df[['Instagram_hrs', 'Facebook_hrs', 'YouTube_hrs']].mean()
fig, ax = plt.subplots()
sns.barplot(x=mean_platform.index, y=mean_platform.values, ax=ax, hue=mean_p
plt.xlabel('Platform')
plt.ylabel('Average Hours')
plt.title('Average Time Spent on Each Social Media Platform')
plt.grid(True,axis='y')
ax.set_axisbelow(True)
plt.tight_layout()
plt.figure()
plt.show()
```

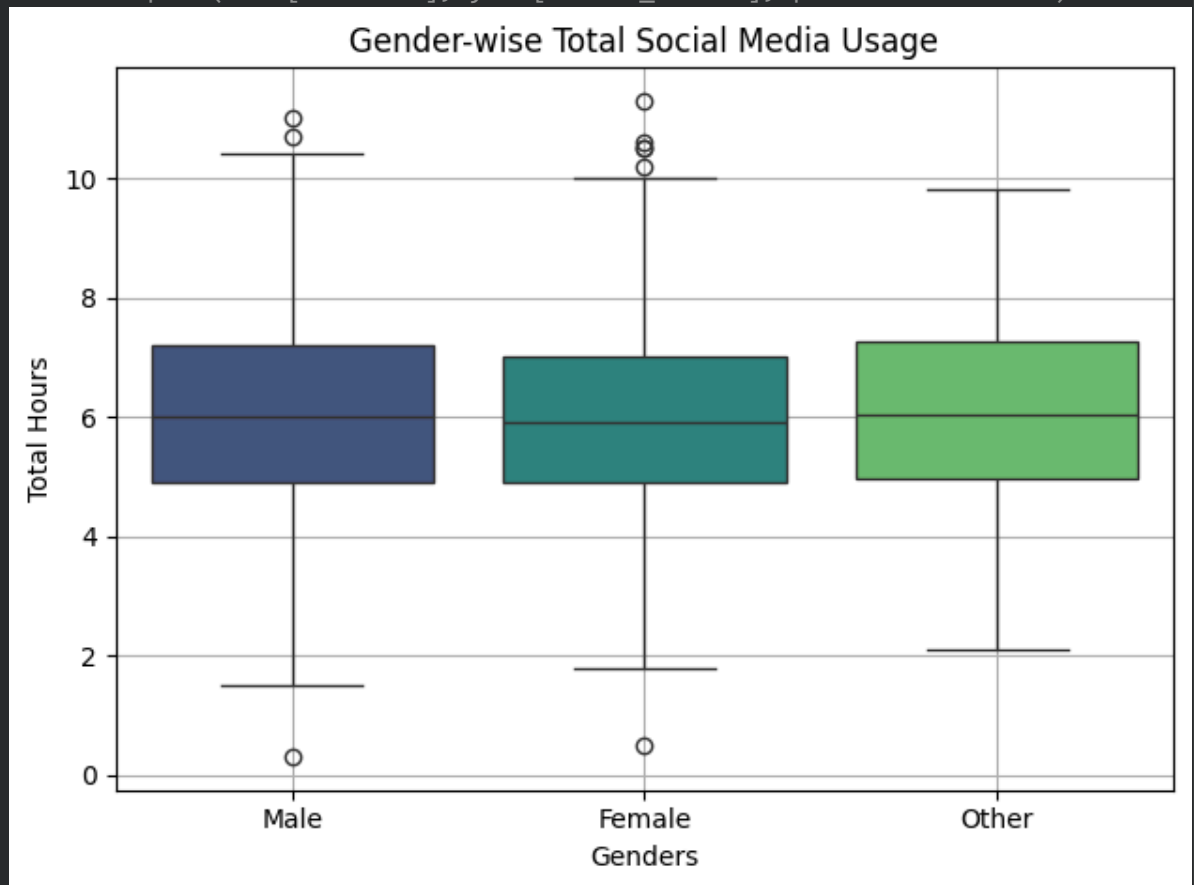


<Figure size 640x480 with 0 Axes>

```
sns.boxplot(x=df['Gender'], y=df['Total_Hours'], palette='viridis')
plt.xlabel('Genders')
plt.ylabel('Total Hours')
plt.title('Gender-wise Total Social Media Usage')
plt.tight_layout()
plt.grid()
plt.figure()
plt.show()
```

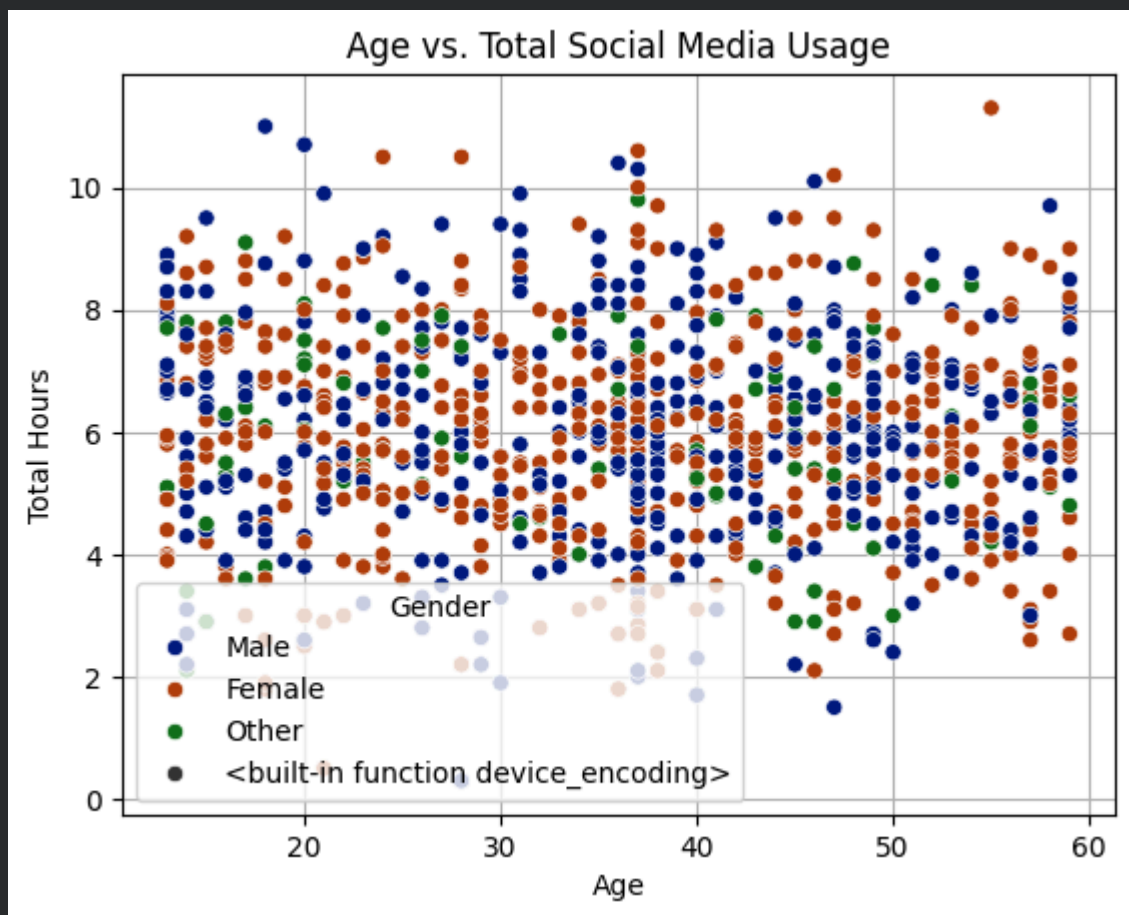
```
/tmp/ipython-input-1038682466.py:1: FutureWarning:
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed in
sns.boxplot(x=df['Gender'], y=df['Total_Hours'], palette='viridis')
```



<Figure size 640x480 with 0 Axes>

```
fig, ax = plt.subplots()
sns.scatterplot(x=df['Age'], y=df['Total_Hours'], ax=ax, hue=df['Gender'], p
plt.grid()
ax.set_axisbelow(True)
plt.xlabel('Age')
plt.ylabel('Total Hours')
plt.title('Age vs. Total Social Media Usage')
plt.figure()
plt.show()
```

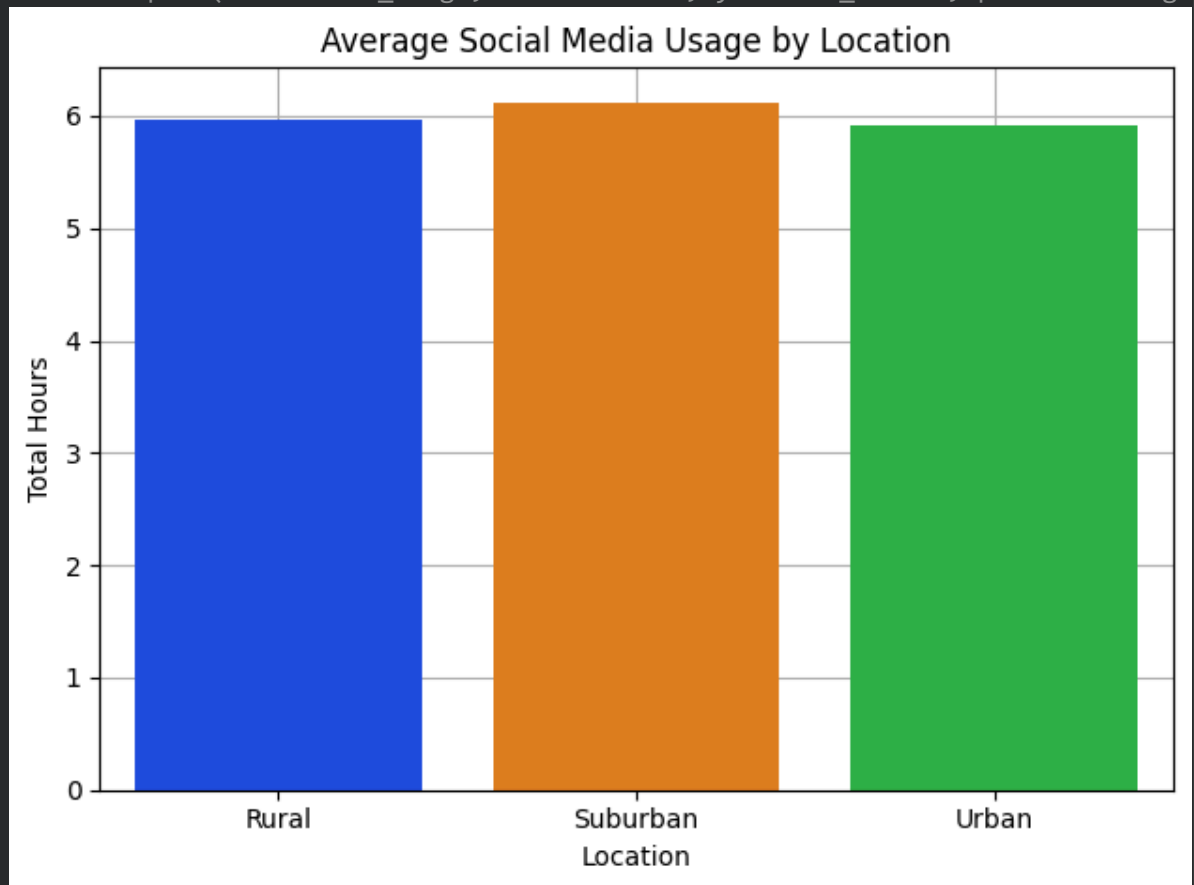


<Figure size 640x480 with 0 Axes>

```
locat_usage = df.groupby('Location')['Total_Hours'].mean().reset_index()
fig, ax = plt.subplots()
sns.barplot(data=locat_usage, x='Location', y='Total_Hours', palette='bright')
plt.title('Average Social Media Usage by Location')
plt.xlabel('Location')
plt.ylabel('Total Hours')
plt.grid(True)
ax.set_axisbelow(True)
plt.tight_layout()
plt.figure()
plt.show()
```

```
/tmp/ipython-input-2660683566.py:3: FutureWarning:
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed i
sns.barplot(data=locat_usage, x='Location', y='Total_Hours', palette='brigh
```



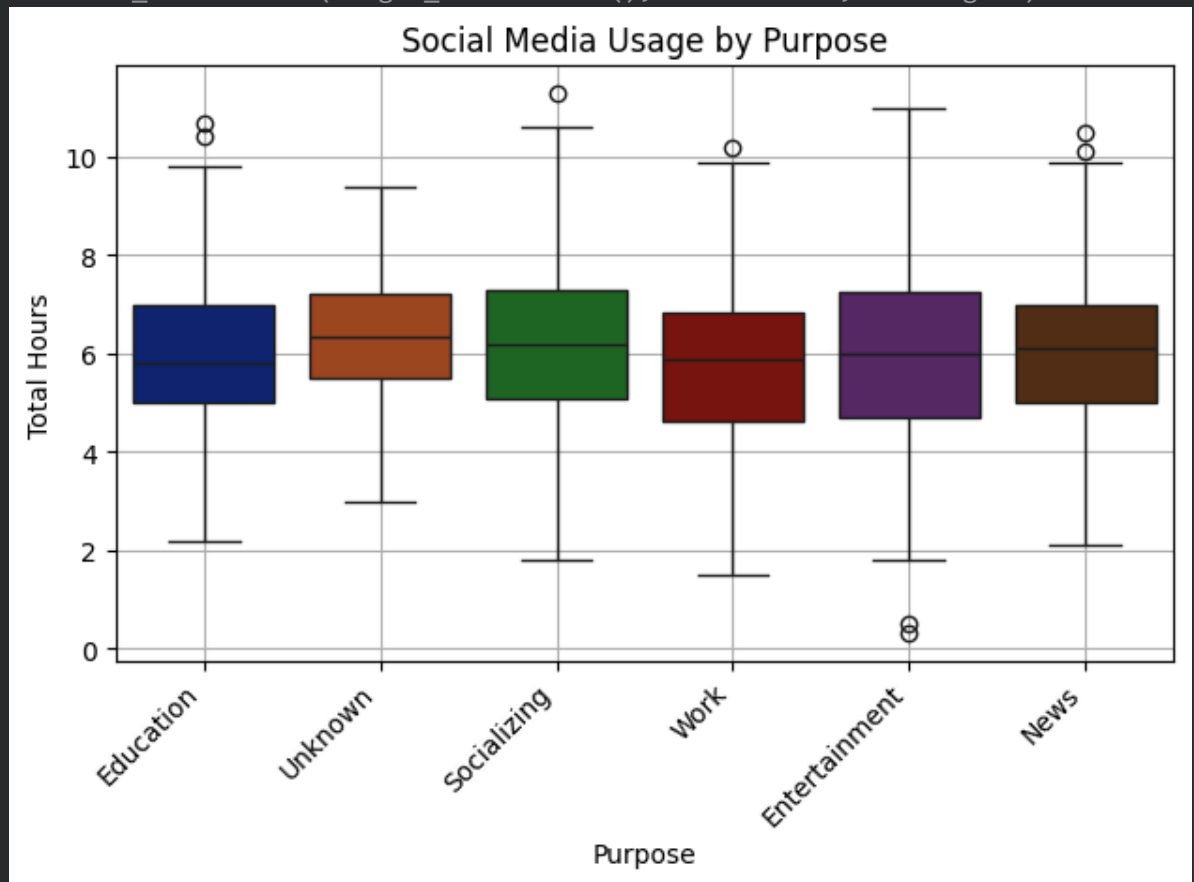
<Figure size 640x480 with 0 Axes>

```
fig,ax = plt.subplots()
sns.boxplot(x=df['Purpose'], y=df['Total_Hours'], ax=ax, palette='dark')
plt.xlabel('Purpose')
plt.ylabel('Total Hours')
plt.title('Social Media Usage by Purpose')
plt.grid()
ax.set_axisbelow(True)
ax.set_xticklabels(ax.get_xticklabels(), rotation=45, ha='right')
plt.tight_layout()
plt.figure()
plt.show()
```

```
/tmp/ipython-input-3368051199.py:2: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in

```
sns.boxplot(x=df['Purpose'], y=df['Total_Hours'], ax=ax, palette='dark')
/tmp/ipython-input-3368051199.py:8: UserWarning: set_ticklabels() should only
ax.set_xticklabels(ax.get_xticklabels(), rotation=45, ha='right')
```

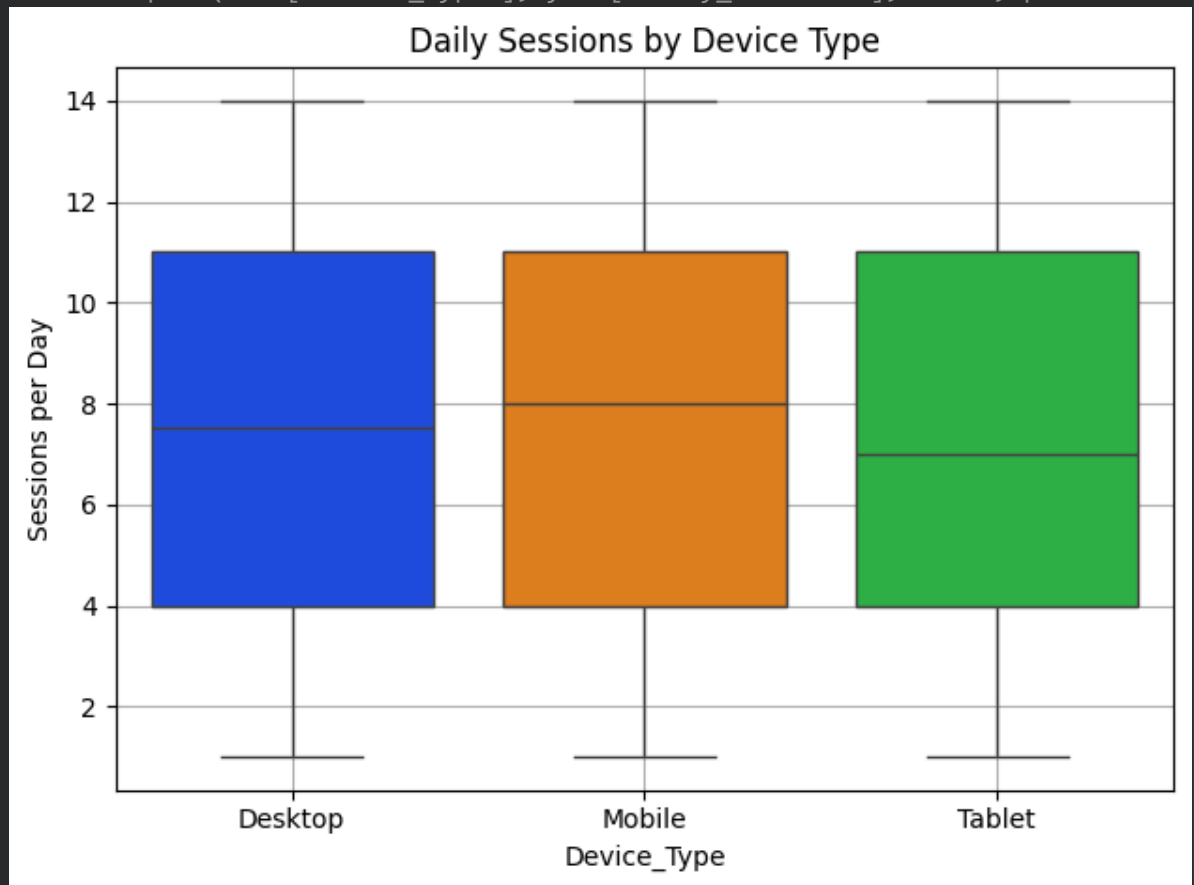


<Figure size 640x480 with 0 Axes>

```
fig,ax = plt.subplots()
sns.boxplot(x=df['Device_Type'], y=df['Daily_Sessions'], ax=ax, palette='bri
plt.xlabel('Device_Type')
plt.ylabel('Sessions per Day')
plt.title('Daily Sessions by Device Type')
plt.grid()
ax.set_axisbelow(True)
plt.tight_layout()
plt.figure()
plt.show()
```

```
/tmp/ipython-input-4104761996.py:2: FutureWarning:
```

```
Passing `palette` without assigning `hue` is deprecated and will be removed i  
sns.boxplot(x=df['Device_Type'], y=df['Daily_Sessions'], ax=ax, palette='br
```



<Figure size 640x480 with 0 Axes>

```
top_user = df.loc[df['Total_Hours'].idxmax()]  
print(top_user)
```

```
User_ID      865  
Age          55.0  
Gender       Female  
Instagram_hrs  4.0  
Facebook_hrs  3.3  
YouTube_hrs  4.0  
Purpose      Socializing  
Location     Rural  
Device_Type  Mobile  
Daily_Sessions  13  
Total_Hours  11.3  
Name: 864, dtype: object
```

```
filter_users = df[df['Age'] < 18]  
young_minimalist = filter_users.loc[filter_users['Total_Hours'].idxmin()]  
print(young_minimalist)
```

```
User_ID      492  
Age          14.0  
Gender       Other  
Instagram_hrs  1.0  
Facebook_hrs -0.0
```

```
YouTube_hrs      1.1
Purpose          Socializing
Location         Suburban
Device_Type      Tablet
Daily_Sessions   2
Total_Hours      2.1
Name: 491, dtype: object
```

```
Engineering College\nProject Week2 Done\nEmail: sanjaykumaran6907@gmail.com")
```

```
Name: Sanjay Kumaran D
College: Vel Tech Multi Tech Dr.Rangarajan Dr.Sakunthala Engineering College
Project Week2 Done
Email: sanjaykumaran6907@gmail.com
```